

Arvin Zaker

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Education

Doctor of Medicine (MD) <i>University of British Columbia, Vancouver, BC</i>	Aug 2025–Present
Honours Bachelor of Science, Translational and Molecular Medicine <i>University of Ottawa, Ottawa, ON; cGPA: 9.98/10</i>	Sep 2019–Apr 2023

Research Experience

Medical Imaging Artificial Intelligence <i>University of British Columbia, Vancouver, BC</i>	Feb 2026–Present <i>Supervisor: Ilker Hacihaliloglu, PhD</i>
- Curated 14 datasets and built a Python pipeline to train 1,176 models on 38,357 paired mammograms. - Externally tested selected models on EMBED dataset to identify AI model performance failures and its association with breast density, race, ethnicity, and histology.	
E-Cigarette Flavour Policy Evaluation <i>University of British Columbia, Vancouver, BC</i>	Feb 2026–Present <i>Supervisor: Kevin Duan, MD</i>
Compared the policy effects of comprehensive e-cigarette flavour bans and specialty-retailer restrictions among 1,516 youth across 5 Canadian provinces using CTNS data and survey-weighted difference-in-differences.	
Epigenomics and Chromatin Biology 🔄 <i>University of Ottawa, Ottawa, ON</i>	Mar 2024–Jan 2026 <i>Supervisor: Dr. Arvind Mer, PhD</i>
- Analyzed clinical exosomal DNA to identify differentially methylated regions and pathways and integrated multi-omics evidence from 44 studies and databases. - Built a reproducible CUT&RUN workflow for processing, quality control, and epigenomic annotation. - Aggregated and visualized CXXC1 mutational data across the genome and plant homeodomain (PHD).	
Text-Mining Feature Selection for Anticancer Drug Response 🔄 <i>University of Ottawa, Ottawa, ON</i>	Sep 2023–Feb 2024 <i>Supervisor: Arvind Mer, PhD</i>
- Devised and benchmarked text-mining feature-selection methods across drug-response prediction models and validation datasets. - Built reproducible high-performance computing workflows and visualizations for model evaluation.	
Bacterial Genomics 🔄 <i>University of Ottawa, Ottawa, ON</i>	Jul 2023–Feb 2024 <i>Supervisors: Arvind Mer, PhD; Vanessa M. D'Costa, PhD</i>
Built genomic annotation and phylogenetic-analysis workflows and characterized type IV secretion system clusters across 118 <i>Acinetobacter</i> bacterial and plasmid strains.	
SMC5/6 Cancer Phenogenomics 🔄 <i>University of Ottawa, Ottawa, ON</i>	May 2022–Aug 2022 <i>Supervisors: Arvind Mer, PhD; Damien D'Amours, PhD</i>
Analyzed SMC5/6 alterations, genomic instability, transcriptomic changes, cancer phenotypes, and survival in a pan-cancer cohort of more than 65,000 patients across 144 studies.	
Optogenetic Neurophysiology <i>University of Ottawa, Ottawa, ON</i>	Mar 2022–Sep 2024 <i>Supervisors: Arvind Mer, PhD; Greg Silasi, PhD</i>
Co-developed software to quantify <i>in vivo</i> hemodynamic and cortical responses to intermittent theta-burst stimulation (iTBS) after stroke.	
Patient-Derived Xenograft Drug-Response Modelling <i>University of Ottawa, Ottawa, ON</i>	Jan 2022–June 2026 <i>Supervisor: Arvind Mer, PhD</i>
- Developed xeMetron package to calculate 250 drug response metrics in patient-derived xenografts. - Built machine-learning models to identify pharmacogenomic biomarkers and quantify drug response in clinical-trial data.	
Bacteriophage Discovery <i>University of Ottawa, Ottawa, ON</i>	Sep 2021–Apr 2022 <i>Supervisor: Adam Rudner, PhD</i>
Isolated, characterized, and annotated the genome of 2 novel bacteriophages, Arzan and Khorshid.	

- Designed electrophysiology experiments on adenosine receptors and skeletal-muscle fatigue and performed tissue staining, fluorescence imaging, and quantitative image analysis.

Peer-Reviewed Publications

1. Asare-Werehene M, **Zaker A**, Tripathi S, Communal L, Carmona E, Mes-Masson A-M, Tsang BK, Mer A. Spotlight on nuclear PD-L1 in ovarian cancer chemoresistance: hidden but mighty. *Front Immunol.* 2025;16. doi:10.3389/fimmu.2025.1543529.
2. Grégoire S, Grégoire J, Yang Y, Capitani S, Joshi M, Sarvan S, **Zaker A**, Ning Z, Figeys D, Ulrich K, Brunzelle JS, Mer A, Couture J-F. Structural insights into an atypical histone binding mechanism by a PHD finger. *Structure.* 2024;32(9):1498–1506.e4. doi:10.1016/j.str.2024.06.017.
3. Wu G, **Zaker A** (co-first author), Ebrahimi A, Tripathi S, Mer AS. Text-mining-based feature selection for anticancer drug response prediction. *Bioinform Adv.* 2024;4(1):vbae047. doi:10.1093/bioadv/vbae047.
4. Nasser F, Gaudreau A, Lubega S, **Zaker A**, Xia X, Mer AS, D'Costa VM. Characterization of the diversity of type IV secretion system-encoding plasmids in *Acinetobacter*. *Emerg Microbes Infect.* 2024;13(1):2320929. doi:10.1080/22221751.2024.2320929.
5. Roy S, **Zaker A**, Mer A, D'Amours D. Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *NAR Cancer.* 2023;5(3):zcad047. doi:10.1093/narcan/zcad047.

Presentations, Posters, Seminars & Abstracts

1. **Poster and oral presentations; presenter.** Roy S, **Zaker A**, Mer A, D'Amours D. Large-scale phenogenomic analysis of human cancers uncovers frequent alterations affecting SMC5/6 complex components in breast cancer. *Canadian Society for Molecular Biosciences International Conference*; May 2023; Ottawa, ON.
2. **Oral presentation; presenter.** **Zaker A**, Mer A. Comparative analysis of drug response metrics in patient-derived xenografts. *4th Annual University of Ottawa Faculty of Medicine Research Day*; Sep 2022; Ottawa, ON.
3. **Abstract and presentation; co-presenter.** Featherstone A, Radar A, Salama A, Tiukuvaara S, Ferri O, **Zaker A**, et al. Give us a millet of your time: dehusking the singleton bacteriophage Arzan. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
4. **Abstract and presentation; co-presenter.** Friesen R, Algharbi S, **Zaker A**, Mahdi O, Gao R, Ferri O, et al. Investigating nucleotide-binding proteins in bacteriophage JohnDoe. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
5. **Abstract and presentation; co-presenter.** Cildir Z, Abdisalam S, Alam A, **Zaker A**, Toex N, Chandra A, et al. Prophage liberation! A method to increase the titer of the temperate phages Khorsid and Winseler. *SEA Symposium*; Apr 2022; Pittsburgh, PA.
6. **Seminar; presenter.** **Zaker A**, Mer A. Comparative analysis of drug response metrics in patient-derived xenografts. *University of Ottawa Translational and Molecular Medicine Seminar*; Apr 2022.
7. **Poster; presenter.** **Zaker A**, Mer A. Analysis of drug response in patient-derived xenografts. *Faculty of Medicine, University of Ottawa*; Feb 2022; Ottawa, ON.
8. **Poster and oral presentations; presenter.** **Zaker A**, McCarthy L, Jung D, Karunakaran G, Rudner A. Discovery and characterization of bacteriophage Arzan. *University of Ottawa Faculty of Medicine*; Dec 2021; Ottawa, ON.
9. **Abstract and poster presentation; presenter.** **Zaker A**, McRae C, Renaud JM. Role of adenosine receptors in mouse skeletal-muscle fatigue kinetics. *University of Ottawa Undergraduate Research Opportunity Program Symposium*; Mar 2021.

Leadership & Governance

Research and Innovation Lead <i>MD Student Learning Group, University of British Columbia, Vancouver, BC</i> Lead the research and innovation portfolio, connecting peers with learning opportunities and promoting evidence-informed practice, emerging technologies, and student-led scholarship.	Jul 2026–Present
Vice-President, External Relations <i>Artificial Intelligence in Medicine Society, University of British Columbia, Vancouver, BC</i> Helped launch a student community focused on responsible artificial intelligence in health care and build external partnerships for panels, workshops, and collaborative projects.	Jul 2026–Present
Chief Technology Officer <i>Learn Freely, nonprofit organization, Vancouver, BC</i> Lead development of a learning management system and workflows for tutoring assignments, progress tracking, project management, and organizational collaboration.	Jun 2026–Present
Board Member <i>Inclusion, Diversity and Equity Advisory Committee, Doctors of BC, Vancouver, BC; three-year term</i> Selected to advise the Board of Directors on equity, diversity, and inclusion across governance bodies, members, and the medical profession.	Jun 2026–Present
Founder and President <i>Hands on Health Magazine, University of Ottawa, Ottawa, ON</i> - Founded and led a 20+ member team that published 50+ faculty-reviewed, evidence-based health articles. - Built editorial, review, and communication workflows that sustained publication during remote operations.	Jun 2020–Sep 2024
Founder and President <i>Canadians Sharing Locally and Globally uOttawa, University of Ottawa, Ottawa, ON</i> - Partnered with a local charity to help raise more than \$100,000 for personal protective equipment and ventilators for rural hospitals in India. - Led advocacy events and social media outreach on global health inequities and social determinants of health.	Dec 2020–Apr 2023

Teaching & Mentorship

Founder and Production Designer <i>UBC Integrated Anatomy Atlas, University of British Columbia, Vancouver, BC</i> - Collaborated with UBC anatomy faculty to develop a comprehensive flashcard deck spanning both preclinical years of the UBC MD curriculum. - Produced anatomy images and 320 structure-identification questions across 14 laboratory modules in Anki, study-site, and simulated bell-ringer formats. - Integrated the resource into UBC Elentra, making it available to all UBC medical students.	Apr 2026–Present
Premedical Student Mentor <i>Supporting Applications, Supporting Interviews (SASI), Community of Support, Temerty Faculty of Medicine, University of Toronto, Toronto, ON</i> Mentored 3 premedical students from groups underrepresented in medicine through one-on-one meetings, individualized application strategy and feedback on interview and written submissions.	Aug 2025–Present
Science, Technology, Engineering, and Mathematics Tutor <i>Oxford Learning Oak Ridges and Inspiration Learning, Toronto, ON</i> Tutored 20+ students from grades 1–12 and university across biology, chemistry, physics, and mathematics using individualized learning plans.	Sep 2023–Jul 2025
Peer Tutor and Mentor Guided 50+ middle-school and high-school students in science and mathematics and supported transitions to university through academic guidance and peer mentorship.	Sep 2018–Present

Community Engagement

First-Year Representative	Sep 2025–Jul 2026
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Persian Medical Society, University of British Columbia, Vancouver, BC

- Help create and promote peer-support, mentorship, and cultural events for Persian-identifying students and allies, strengthening community connection and inclusive representation within the medical school.

Administrative Receptionist	Sep 2022–Jun 2023
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Chadha Medical Clinic, Ottawa, ON

- Coordinated appointments, referrals, and screenings in a primary-care practice serving approximately 5,000 patients and managed 60+ patient contacts weekly.

Volunteer Coordinator and Co-Chair	Sep 2020–Sep 2024
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Best Buddies uOttawa, University of Ottawa, Ottawa, ON

- Recruited and paired 50+ student volunteers with people with intellectual and developmental disabilities.
- Led accessible events and revised recruitment to address audiovisual and language barriers identified through feedback.

Honours, Scholarships & Awards

Florence E. Heighway Summer Research Award	Jun 2026–Aug 2026
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Faculty of Medicine Summer Student Research Program (SSRP), University of British Columbia, Vancouver, BC

- Awarded 8 weeks of funding for a mammography artificial-intelligence failure-analysis project.

Third Place, IFHP NavAID Challenge	Jun 2026
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University of British Columbia, Vancouver, BC

- Co-developed a 75-minute simulation module aimed at training health professionals to navigate Canada's Interim Federal Health Program and coordinate interprofessional care for refugees.

Undergraduate Student Research Award (USRA)	May 2023–Aug 2023
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Natural Sciences and Engineering Research Council of Canada (NSERC), University of Ottawa, Ottawa, ON

- One of the 88 recipients; awarded \$7,500 for a 3-month full-time research project based on academic merit and proposal quality.

Translational and Molecular Medicine Summer Training Award	May 2022–Aug 2022
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University of Ottawa, Ottawa, ON

- Awarded \$8,900 for a 3-month full-time research project based on academic merit and proposal quality.

Work in Biomedical Research Scholarship	May 2022–Aug 2022
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University of Ottawa, Ottawa, ON

- Awarded \$3,500 for a 3-month full-time research project through a competition with a success rate below 50%.

Undergraduate Research Opportunity Program Award	Dec 2020
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University of Ottawa, Ottawa, ON

- Awarded \$1,000 for a 4-month supervised research project based on academic merit and proposal quality.

Dean's Honour List and Merit Scholarship	Sep 2019–Apr 2023
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University of Ottawa, Ottawa, ON

Projects & Interests

Explainable Heart-Disease Risk Modelling

Mar 2026

National Canadian Medical Datathon, Vancouver, BC

- Collaborated with 5 colleagues to integrate 920 records from 4 University of California, Irvine heart-disease cohorts and develop a regularized logistic-regression model.
- Built CLEAR, an R Shiny demonstration tool providing explainable and individualized risk estimates using SHapley Additive exPlanations and plain-language feature summaries.

Clinical Bioinformatics Reproducibility Project

Mar 2023

- Reproduced the bioinformatics analysis reported by Viñas et al. (2020; PMID: 32662516) and produced a documented report, presentation, and explanatory visualizations.

Systems Administration and Research Computing

Jan 2022–Present

- Built and maintain a 16-terabyte Unraid server for centralized family storage, automated backups, data recovery, and Docker-based web services.
- Built and administer 3 Linux workstations for machine learning, and high-performance computing.